

Project Report: ScamGuard

A Rule-Based Social Engineering and Scam Detection Tool

EXECUTIVE SUMMARY

As a cyber warrior, I frequently encounter various problems regarding scam and fraud messages affecting everyday digital users. One of the most common challenges people face is the inability to suspect, identify, and verify whether an incoming message is legitimate or fraudulent. Scammers heavily rely on psychological manipulation—creating artificial urgency, threatening account suspensions, or offering fake rewards—to force panic-driven decisions.

With the primary motive to overcome this problem and protect everyday users, I designed **ScamGuard**, a social engineering and scam detection application. This tool helps users identify scam and fraud messages by analyzing their underlying text patterns, psychological urgency cues, credential-harvesting indicators, and unverified external links to calculate a clear, dynamic risk score.

1. PROBLEM STATEMENT

Digital communications such as SMS, instant messaging, and emails have become primary vectors for phishing and social engineering attacks. Non-technical users often lack the security awareness required to detect subtle threat indicators. Key challenges include:

- **Psychological Exploitation:** Messages engineered to induce fear, urgency, or excitement.
- **Credential Theft:** Impersonation of legitimate institutions (e.g., banks, courier services) to harvest One-Time Passwords (OTPs), PINs, and passwords.
- **Deceptive Links:** Use of unverified URLs and shortened domains designed to redirect users to malicious landing pages.

2. PROJECT OVERVIEW & SOLUTION

ScamGuard provides an accessible, lightweight line of defense. Built with an intuitive desktop interface, the application processes suspicious raw text inputs provided by the user, normalizes the content, and executes a rule-based evaluation.

Core Features

- **Instant Risk Scoring:** Evaluates text and generates a threat score from 0 to 100.
- **Visual Categorization:**
 -  **Low Risk (0–39):** Minimal to no suspicious patterns identified.

-  **Medium Risk (40–69):** Moderate risk indicators present; exercise caution.
-  **High Risk (70–100):** Multiple critical threat vectors detected (e.g., requests for sensitive credentials paired with urgency).
- **Actionable Advice:** Delivers immediate, practical safety guidelines tailored to the evaluated risk level.
- **Privacy-Centric Architecture:** Runs completely offline, ensuring user data and private communications are never transmitted to external servers.

3. METHODOLOGY & EVALUATION ENGINE

The detection engine operates deterministically through pattern-matching logic implemented via regular expressions and string analysis. Each detected threat vector contributes a specific weight to the aggregate risk score:

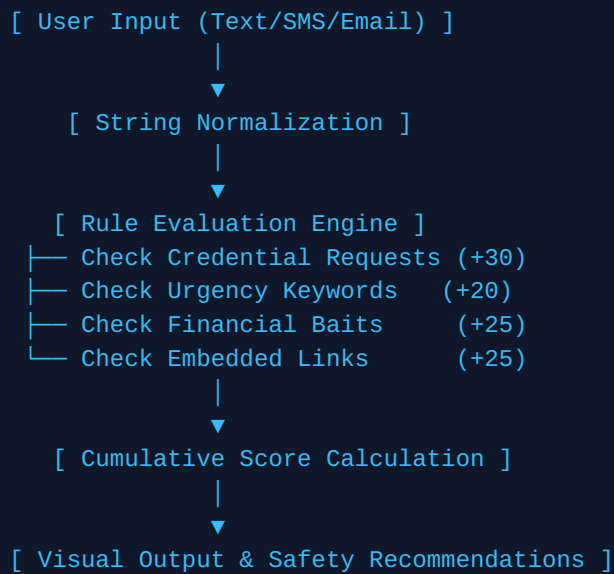
THREAT VECTOR	ANALYSIS CRITERIA	SCORE IMPACT
Credential Requests	Detects requests for sensitive data such as OTPs, banking PINs, passwords, and personal identifiers.	+30 Points
Urgency Keywords	Flags psychological triggers like "Act Immediately", "Suspended", "Urgent", or "Account Blocked".	+20 Points
Financial / Prize Baits	Identifies fake rewards, lottery claims, cash prize notifications, or refund offers.	+25 Points
External Link Analysis	Scans for embedded URLs, HTTP protocols, and unverified external web domains.	+25 Points

4. SYSTEM ARCHITECTURE & TECH STACK

Programming Language: Python 3

Graphical User Interface (GUI): Tkinter

Pattern Matching: Python re module (Regular Expressions)



5. FUTURE SCOPE

- **Machine Learning Integration:** Upgrading the rule-based engine with Natural Language Processing (NLP) models (e.g., Naïve Bayes or BERT) to detect context-based context manipulation.
- **URL Reputation APIs:** Integrating real-time threat intelligence feeds to check embedded URLs against active phishing databases.
- **Multi-Language Detection:** Expanding regex and model parameters to detect localized scam attempts in regional languages.

6. CONCLUSION

ScamGuard addresses a crucial gap in user-end cyber defense by converting complex threat indicators into a clear, quantifiable score. By empowering individuals to evaluate suspicious messages independently and offline, the application reduces susceptibility to social engineering attacks and promotes safer digital habits.